THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(c)** 105	Complete graph on n points: $\frac{n(n-1)}{2}$.
2	**(c)** 150	Find the new time first, then the angle: each hour mark is 30°.
3	**(d)** 5	Repeat the largest-minus-smallest step.
4	**(b)** 140	Total minus the largest.
5	**(a)** 1001	Divide 1000 by the number and round up.
6	**(c)** 162	Let breadth $= b$.
7	**(c)** $\frac{23}{24}$	Use the LCM of the denominators.
8	**(d)** 2	
9	**(c)** 5	Number of sides $= 360 \div a$.
10	**(d)** 22	Simplify step by step.
11	**(c)** 37	Try triangular numbers.
12	**(d)** 1170	$360n + 90$.
13	**(b)** 24	List the sequence.
14	**(d)** 4	Divide by 3 (the symbols are in the same proportion).
15	**(d)** 19	Total numbers − primes − 1 (1 is neither).
16	**(b)** 16	Perimeters are equal.
17	**(d)** 60	Difference of the fractions, then divide.
18	**(c)** 20	Add the three sides.
19	**(c)** 45	Divide.
20	**(a)** 30	Opposites cancel.
21	**(c)** 800	Subtract the first 10 odd numbers, whose sum is 100.
22	**(b)** 180	Net turn.
23	**(c)** 10	The second digit can be any of 10 digits.
24	**(b)** 84	Multiply.
25	**(b)** 24	LCM.
26	**(c)** 120	Half of the rectangle.
27	**(b)** 40	Fraction removed = difference.
28	**(b)** 36	Side $= 2r$.
29	**(b)** 2	Letters H, I, O, X.
30	**(b)** 23	Distance $= \lvert a - b \rvert$.
31	**(b)** 39	$20^2 - 19^2$ is the next odd number.
32	**(a)** 30	The hands are placed symmetrically about the 12-6 line, so the angle is the same.
33	**(a)** 165	Add the numbers.
34	**(b)** 9	Add the children, then divide by the value of one symbol.
35	**(c)** 24	HCF of three numbers.
36	**(a)** 336	Outer minus inner.
37	**(c)** 8	Divide the length by the piece length.
38	**(a)** 16	Two diameters.
39	**(c)** 150	Add the smallest angles.
40	**(b)** -17	Subtract.

41	**(a)** Both A and R are true and R is the correct explanation of A.	15 + 4 = 19, and R is the rule of the pattern.
42	**(a)** Both A and R are true and R is the correct explanation of A.	135 lies between 90 and 180, and R is exactly the definition.
43	**(a)** Both A and R are true and R is the correct explanation of A.	7, 5, 3, 0 in descending order gives 7530.
44	**(a)** Both A and R are true and R is the correct explanation of A.	3.5 x 10 = 35, and R says what the half symbol means.
45	**(a)** Both A and R are true and R is the correct explanation of A.	A prime has exactly two factors and a composite has more than two; 1 has just one factor.
46	**(a)** Both A and R are true and R is the correct explanation of A.	2 x (12 + 5) = 34.
47	**(a)** Both A and R are true and R is the correct explanation of A.	Multiplying 3 and 4 by 2 gives 6 and 8.
48	**(a)** Both A and R are true and R is the correct explanation of A.	Every chord through the centre has length 2r, and no other chord is longer.
49	**(a)** Both A and R are true and R is the correct explanation of A.	A square is a regular polygon with 4 sides.
50	**(a)** Both A and R are true and R is the correct explanation of A.	-7 lies to the left of -3.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers — 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(d)** 88	Add the triangular number.
2	**(b)** 4	Each line gives one straight angle.
3	**(b)** 4	Find the palindrome.
4	**(b)** 16	Add the heights.
5	**(b)** 997	Subtract the remainder first.
6	**(c)** 5	Area = cost ÷ rate; breadth = area ÷ length.
7	**(c)** $\frac{7}{10}$	Compare with a common denominator.
8	**(b)** 18	Each side is $2r$.
9	**(a)** 3	Order n gives $n-1$ such turns before a full turn.
10	**(c)** -6	The first two cancel.
11	**(a)** 2025	Square of the triangular number.
12	**(c)** 72	$360^\circ \div 5$.
13	**(d)** 968	Reverse and add.
14	**(d)** 8	Integer division.
15	**(a)** 41	List the primes after 10.
16	**(b)** 64	A square has the greatest area for a given perimeter.
17	**(c)** $2\frac{3}{4}$	Subtract.
18	**(c)** 70	Base angles are equal.
19	**(c)** 9	$360\div$ angle.
20	**(c)** -5	Add b to both sides.
21	**(a)** 41	Each triangular number adds the next counting number.
22	**(b)** 150	Multiply the angle between neighbours by the number of gaps.
23	**(c)** 3	Continue the process.
24	**(b)** 7.5	Divide by the scale.
25	**(b)** 97	Check numbers just below.
26	**(a)** 42	Add the three sides.
27	**(d)** 24	LCM.
28	**(a)** 10	Diameter equals the side.
29	**(d)** 42	Number of sides times side.
30	**(a)** 2	Subtracting a negative adds.
31	**(d)** 153	Complete graph on n points: $\frac{n(n-1)}{2}$.
32	**(a)** 90	Find the new time first, then the angle: each hour mark is 30°.
33	**(c)** 7	Repeat the largest-minus-smallest step.
34	**(d)** 160	Total minus the largest.
35	**(a)** 6	$330 = 2 \times 3 \times 5 \times 11$
36	**(b)** 84	Outer side $= a + 2$.
37	**(b)** $1\frac{1}{6}$	Use the LCM of the denominators.
38	**(a)** 2	
39	**(b)** 10	Equal distance on the other side.
40	**(b)** 7	All pairs cancel except the last.
41	**(b)** Both A and R are true but R is not the correct explanation of A.	1+3+5+7+9 = 25, and 25 = 5 x 5 is a square, but R does not explain why the sum is 25.

42	**(b)** Both A and R are true but R is not the correct explanation of A.	Both statements are true, but the value of a right angle does not explain the value of a straight angle.
43	**(a)** Both A and R are true and R is the correct explanation of A.	7 is odd: 3 x 7 + 1 = 22.
44	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true: equal widths keep the graph fair, heights show the data, but R does not explain A.
45	**(a)** Both A and R are true and R is the correct explanation of A.	So every even number above 2 has at least three factors and is composite.
46	**(b)** Both A and R are true but R is not the correct explanation of A.	7 x 7 = 49 and area is in square units, but R does not explain the value 49.
47	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true. R is just another way of writing the number and does not explain why it is improper.
48	**(c)** A is true but R is false.	Radius = 14 / 2 = 7, so A is true. The diameter is twice the radius, so R is false.
49	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true but the order of rotation does not explain the number of lines.
50	**(b)** Both A and R are true but R is not the correct explanation of A.	Both are true, but R does not explain why 5 + (-5) = 0.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–10) | Class 6 Maths | Maximum Marks: 50

MULTIPLE CHOICE QUESTIONS — Answers 50 × 1 = 50

Q.	Correct Option	Brief Reasoning
1	**(a)** 1000	Subtract the first 15 odd numbers, whose sum is 225.
2	**(c)** 270	Net turn.
3	**(d)** 10	The second digit can be any of 10 digits.
4	**(b)** 90	Multiply.
5	**(b)** 34	Total numbers − primes − 1 (1 is neither).
6	**(b)** 20	Perimeters are equal.
7	**(c)** 48	Fraction removed = difference.
8	**(d)** 100	Side $= 2r$.
9	**(b)** 40	Divide.
10	**(b)** 42	Opposites cancel.
11	**(c)** 49	$25^2 - 24^2$ is the next odd number.
12	**(c)** 60	The hands are placed symmetrically about the 12-6 line, so the angle is the same.
13	**(d)** 228	Add the numbers.
14	**(c)** 26	Add the children, then divide by the value of one symbol.
15	**(c)** 60	LCM.
16	**(d)** 240	Half of the rectangle.
17	**(c)** 15	Divide the length by the piece length.
18	**(d)** 24	Two diameters.
19	**(c)** 2	Letters H, I, O, X.
20	**(a)** 16	Distance $= \lvert a - b \rvert$.
21	**(b)** 13	Divide the total increase by the number of steps, which is one less than the term number.
22	**(c)** 40	The minute hand turns 6° per minute.
23	**(d)** 24	Subtract.
24	**(c)** 51	Subtract.
25	**(b)** 30	HCF of three numbers.
26	**(b)** 584	Outer minus inner.
27	**(b)** $10\frac{1}{2}$	Find the number first.
28	**(d)** 24	Each is half the diagonal.
29	**(b)** 135	Add the smallest angles.
30	**(b)** 17	Subtract.
31	**(b)** 145	Add the triangular number.
32	**(d)** 5	Each line gives one straight angle.
33	**(a)** 3	Find the palindrome.
34	**(c)** 23	Add the heights.
35	**(d)** 2	Digit sum must be a multiple of 9.
36	**(d)** 18	Perimeter does not change.
37	**(d)** $\frac{11}{15}$	Compare with a common denominator.
38	**(b)** 30	Each side is $2r$.
39	**(c)** 5	If a figure has a line and order n, it has n lines.
40	**(c)** -110	Gains positive, losses negative.

41	**(c)** A is true but R is false.	1+2+...+10 = 10 x 11 / 2 = 55, so A is true. The sum of the first n natural numbers is $\frac{n(n+1)}{2}$, not $n \times n$, so R is false.
42	**(c)** A is true but R is false.	It turns 360° in 60 minutes, i.e. 6° per minute, so $15 \times 6 = 90^\circ$. A is true, R is false.
43	**(c)** A is true but R is false.	A is true (5432 - 2345 = 3087). A Kaprekar step subtracts the smallest from the largest arrangement, so R is false.
44	**(c)** A is true but R is false.	4 x 20 = 80, so A is true. You multiply the height by the scale value, so R is false.
45	**(c)** A is true but R is false.	91 = 7 x 13 so A is true. 91 is not divisible by 2, 3 or 5 but is composite, so R is false.
46	**(c)** A is true but R is false.	Breadth = 48 / 8 = 6, so A is true. Breadth = area / length, so R is false.
47	**(c)** A is true but R is false.	A is true; the denominator stays the same, so R is false.
48	**(d)** A is false but R is true.	5 > 4, so the point lies outside and A is false. R is true.
49	**(c)** A is true but R is false.	It fits onto itself after a half turn, so A is true. It has no line of symmetry, so R is false.
50	**(c)** A is true but R is false.	-4 + 6 = 2 is true. Subtracting -3 is the same as adding 3, so R is false.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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